Watching a 3.8-billion-parameter transformer think: a browser-native, reference-validated, 1:1 real-time visualization of an LLM forward pass

Ahmet Barış Günaydın *Independent researcher · github.com/abgnydn/neuropulse · neuropulse.live*

Abstract

Existing visualizations of transformer language models fall into two disjoint camps: pedagogical tools that animate *toy* models (a few thousand parameters, a sort-the-alphabet task) in real 3D, and inference engines that run real multi-billion-parameter models with *zero* internal visibility. Nothing connects them — you can either watch a model that isn't real, or run a real model you can't watch. **neuropulse** closes that gap: it renders a real forward pass of Phi-3-mini-4k-instruct (3.8 B parameters, 32 layers) in a web browser, in real time, with every rendered element bound 1:1 to a named tensor in the model's compute graph and read back from the same WebGPU buffers the model just wrote — no interpolation, no sampling, no schematic stand-in, no server. To our knowledge this is the first visualization that is simultaneously (i) a real multi-billion-parameter model, (ii) running in-browser, (iii) in 3D, (iv) over *all* live tensors rather than attention alone, and (v) validated against a reference implementation. The inference engine is hand-written: eleven WGSL compute kernels over MLC's `q4f16_1` weights, 292 GPU dispatches per token on the headless path and 348 with the visualizer reading back per-layer attention scores and eight logit-lens probes. Correctness is not asserted but *tested*: a built-in suite diffs the live WebGPU forward pass against a pinned HuggingFace fp16 reference at nine layer checkpoints and holds per-layer relative L2 $< 2\times10^{-2}$, cosine > 0.999, **100% top-1 token agreement**, and softmax JSD $< 5\times10^{-3}$ — the user's own GPU is the test rig. The platform is built as an empirical lab with a pre-registered prediction log; we demonstrate what that enables with experiment E45, a falsification of a continuous-attention self-consistency hypothesis: iterating Phi-3-mini's attention to its Picard fixed point destroys language coherence at a sharp cliff (iteration 2, universal across 16/16 prompts) and converges to a degenerate low-dimensional attractor, while a well-trained 4-layer control model stays robust (91% top-1 retained) — locating the brittleness in scale and training breadth, not in transformers per se.

---

1. Introduction

Most "AI visualizations" are decoration: particle systems and pulsing dots with no model behind them. They show how a designer *imagines* an LLM works. A smaller set of serious tools show something real, but each gives up a dimension that matters. Bycroft's LLM-Viz renders a transformer in beautiful 3D with live tensors — of a ~1,000-parameter toy that sorts letters. Transformer Explainer runs a real GPT-2-small (124 M) in the browser, in 2D, with partial tensor exposure. BertViz exposes real attention from real HuggingFace models, but only attention, only in a notebook, only in 2D. WebLLM runs real multi-billion- parameter models in the browser at full speed — and shows you nothing inside.

The gap is structural: visualization tools scale *down* to stay legible; inference engines treat internals as opaque to stay fast. neuropulse takes the position that the internals of a *full-scale* model, drawn faithfully, are worth the engineering to expose — both as pedagogy (you can watch a real 3.8 B transformer generate a token) and as an interpretability substrate (every tensor is addressable, so experiments can be run *on the thing being shown*). The contribution is not a new model or a new rendering technique; it is the demonstration that a real, full-scale forward pass can be rendered 1:1 in a browser tab, that the 1:1 claim can be made falsifiable against a reference, and that the resulting apparatus is a usable lab. We show the last point with a pre-registered falsification (§3.3).

2. Methods

2.1 Inference engine

The model is Phi-3-mini-4k-instruct (3.8 B parameters; 32 layers; 32 attention heads, no GQA; head dimension 96; hidden dimension 3,072; SwiGLU FFN inner dimension 8,192; vocabulary 32,064), loaded in MLC's `q4f16_1` ndarray-cache format (4-bit weights, fp16 scales, group size 32) — *not* GGUF. The decode loop is hand-written as eleven WGSL compute kernels (`embedding`, `int4_matmul`, `int4_matmul_f32`, `rope`, `attention`, `attention_scores`, `kv_append`, `rms_norm`, `add_norm`, `fused_ffn`, `argmax`) with no inference framework in the path. One generated token is 292 GPU dispatches on the headless fast path; the visualized path adds 56 (one attention-scores readback per layer plus eight logit-lens probes of three dispatches each) for 348. Weights are streamed once and cached in OPFS and the Cache API for instant reload.

2.2 1:1 rendering by shared-buffer readback

Inference and visualization **share the same GPU buffers**. The renderer does not recompute, approximate, or re-simulate anything: it reads the values the model already produced and maps each to a scene element bound to a named tensor — the residual stream, the 32×32 attention-head grid, the gated MLP, the token strip, the per-layer logit-lens. Spatial layout is derived from the model rather than chosen by a designer: residual-stream positions come from a PCA of the model's own layer-0 `qkv_proj` weight matrix, so dimensions read into attention together cluster together. Ten draggable panels each surface one live tensor (top-k, confidence, KV-cache occupancy, residual norm, per-layer delta, last-layer attention, logit-lens, …).

2.3 Validation methodology

"Strict 1:1" is a strong claim, so it is made falsifiable. A built-in suite diffs the live WebGPU forward pass against a pinned HuggingFace fp16 Phi-3-mini at nine layer checkpoints (`VALIDATE_LAYERS = {0,4,8,12,16,20,24,28,31}`, 27,648 floats compared per prompt). The single intentional precision tradeoff vs the FP32 reference is f16 accumulation of intermediate residuals at every Add+Norm boundary; matmul, softmax, RMSNorm, RoPE, and SiLU all accumulate at f32 inside the kernels. The suite runs on the user's own GPU in under a minute and prints to their console; a runtime fingerprint (build SHA, GPU vendor/architecture, browser) stamps every report. A second, commit-time gate (`verify-claims.mjs`, `check-shortcuts.mjs`) cross-checks every numeric claim in the documentation (layer count, kernel count, dispatch counts, keyboard shortcuts) against the source, so prose cannot silently drift from code.

3. Results

3.1 The visualization system

neuropulse renders a real Phi-3-mini forward pass at interactive rates in a browser tab: type a prompt, and 3.8 B parameters process it with every intermediate tensor read back and drawn. Five view modes (Journey, Scene, Attention, Logit-Lens, Cinematic) and four stacking overlays present the same underlying activations from different angles; the camera, panels, and a keyboard-driven cinematic flythrough cover the interaction surface. Because the render path reads the inference buffers directly, what is on screen *is* the model's state, not a depiction of it.

3.2 Correctness against a reference

The 1:1 claim holds against HuggingFace fp16 within the declared tolerances. The bounds the suite asserts (a run exceeding any is flagged a regression):

Test	Quantity	Bound
Hidden state vs HF	relative L2 per layer (3,072 dims)	$< 2\times10^{-2}$ (observed max $\approx 1.4\times10^{-2}$)
Hidden state vs HF	cosine similarity per layer	> 0.999
Last-attention reconstruction	rel. L2 of (Σ scores·V – attn_out)	$< 1\times10^{-2}$
Top-1 token agreement	greedy argmax, ≤ 30 tokens	**100% match**
Top-5 overlap	mean number of shared top-5 tokens (GPU vs HF)	≥ 4
Softmax divergence	mean JSD over top-k	$< 5\times10^{-3}$
Sampler self-test	JSD(5,000 samples, softmax)	$< 1\times10^{-2}$

The deltas at hidden-state level are the cost of int4 quantization and f16 residual round-trips, not implementation drift; the bar that matters — identical top-1 tokens vs the fp16 reference on the validation prompt set — is met. Each kernel additionally carries a declared per-kernel error budget (e.g. `int4_matmul` ≤ 4 ULP@f16, `rope` ≤ 1 ULP, `kv_append/embedding` bit-exact); wiring the per-kernel CPU references into CI is a tracked gap (§4).

3.3 What the lab produces: experiment E45

The platform exposes every tensor and ships a pre-registered prediction log, so hypotheses can be tested on the model being visualized. Prediction P-20260526-07 asked whether iterating Phi-3-mini's attention to a per-layer self-consistency (Picard) fixed point — `Q ← softmax(QKᵀ/√d)V`, RoPE applied once, K/V held fixed — yields hidden states close to the standard one-step operator. The fixed-point kernel ships behind a flag (`?attn=fixedpoint`); a wiring gate confirmed `max_iter = 1` reproduces the baseline byte-for-byte before any claim was made.

The hypothesis is **falsified at zero-shot inference**. One extra Picard step destroys coherence at a sharp cliff: on "The capital of Japan is …", iteration 1 matches the standard kernel exactly ("…Tokyo. Tokyo is a major global"), iteration 2 collapses ("lee\n\n\nTop \nTop"), and iterations 3–100 converge byte-for-byte to a degenerate topic-projector attractor. A 16-prompt sweep (English, code, math, Japanese, emoji, JSON) makes the cliff universal: **16/16 prompts that are coherent at iteration 1 collapse at iteration 2**, with a low-dimensional attractor vocabulary (`"lee"`, `"ício"`, `"RESS"`, newlines) and a second failure mode (immediate stop-token / empty output) on two prompts. The Picard fixed point demonstrably *exists*, is numerically reachable (‖Q_t – Q_{t-1}‖_∞ falls to f32 noise within 100 iterations), and costs only 1.41× compute at `max_iter = 100` — so the cliff is the science, not an artifact of under-iteration or budget.

A disentanglement control separates scale from the phenomenon: a well-trained 4-layer / 64-hidden toy transformer (same RMSNorm + SwiGLU + RoPE family) under the identical protocol stays in the "match" bucket — 91% top-1 retained at iteration 2, stable through iteration 100 — where Phi-3-mini retains 0% coherent output. The publishable claim is therefore stronger than "Path A falsified": **a trained discrete-attention layer's distance from its own Picard fixed point grows with model scale and training breadth; brittleness to attention iteration is a property of well-trained large LMs specifically, not of transformers in general.** A per-layer single-layer-ablation (Phase 3) to adjudicate the two pre-registered per-layer hypotheses is the natural follow-up and is not yet run.

4. Honest limitations

- **Per-kernel error budgets are declared, not yet CI-asserted.** The bounds in §3.2 are documented and the end-to-end HF parity is enforced on every user click, but the runner that checks each kernel against a CPU f64 reference does not yet exist.
- **Cross-vendor parity is under-sampled.** The validation suite has been run on Apple M-series and a single NVIDIA workstation; the cross-vendor matrix is open and users are asked to report fingerprint + parity numbers.

- **One model.** The engine is specialized to Phi-3-mini's architecture and the MLC `q4f16_1` format (whose RoPE base/theta-scaling we inherit). Generalizing to other architectures is engineering not yet done.
- **Not a training or fine-tuning tool**, and no claim is made about the model's fairness, alignment, or safety — only that what is rendered faithfully matches the reference forward pass.
- **E45 per-layer attribution is deferred.** The output cliff at iteration 2 is sharp enough that the two pre-registered per-layer hypotheses (deepest-middle vs high-entropy layers as the catastrophic ones) are both partially refuted at model-level granularity; the single-layer ablation that would pick a winner is Phase 3, unrun.

5. Related work

Prior interactive transformer visualizations trade scale for visibility or vice versa: Bycroft's LLM-Viz (real 3D, live tensors, ~1 K-parameter toy), Transformer Explainer (real GPT-2-small, 124 M, 2D, partial tensors), BertViz (real attention only, 2D, notebook), and inference runtimes such as WebLLM (real multi-billion-parameter models in-browser, no internal visibility). Across the axes *real model · multi-billion scale · browser · 3D · all live tensors · reference-validated*, neuropulse is, to our knowledge, the first to satisfy all six simultaneously. The interpretability result in §3.3 relates to fixed-point / equilibrium views of attention (deep equilibrium models; Ramsauer et al.'s modern Hopfield reading of attention); we test the *inference-time* self-consistency point of a trained discrete-attention model and find it qualitatively unlike the one-step operator, with a scale-dependent severity made visible by the control.

6. Software availability and reproducibility

Source: `github.com/abgnydn/neuropulse` (MIT); live at `neuropulse.live`. The WGSL kernels are in `src/engine/shaders/`; the canonical architecture constants and derived dispatch counts in `src/engine/compiler.ts` and `src/engine/phi3-facts.ts`; precision matrix, tolerances, and per-kernel budgets in `METHODS.md`; the pre-registered prediction log in `PREDICTIONS.md`. The HF parity suite runs on any WebGPU-capable Chromium from the demo's validate button and regenerates its reference via `tools/dump_phi3_reference.py`; the E45 artifacts are committed under `tests/results/`. Documentation numbers are gated against source by `npm run verify`. The Butterfly context-compaction study that also ran on this platform has its own repository (`github.com/abgnydn/butterfly`).

Generative-AI disclosure. Portions of the software, documentation, and this manuscript were drafted with a large language model used as a coding and writing aid; all output was author-reviewed, correctness was enforced by the HF parity suite and the commit-time claim checks, and every quantitative claim is traceable to `METHODS.md`, `phi3-facts.ts`, or a committed experiment artifact.

Statements. Sole author; no competing interests; no external funding. Model weights are Microsoft's Phi-3-mini redistributed by MLC and are not part of this work.

7. Conclusion

A full-scale transformer forward pass can be rendered 1:1, in real time, in a browser tab — every element a real tensor read from the buffers the model just wrote, every "1:1" claim falsifiable against a HuggingFace reference the user runs on their own GPU. Closing the gap between toy-but-visible and real-but-opaque yields both a pedagogical artifact (watch 3.8 B parameters produce a token) and an interpretability substrate: experiment E45 used it to falsify a continuous-attention self-consistency hypothesis and, via a trained small-model control, to attribute the resulting brittleness to scale and training breadth rather than to transformers in general. The platform's discipline — derived constants, a pre-registered prediction log, and a validation suite that fails loud — is the point as much as the rendering is.

References (to be formatted)

- **Bycroft** — LLM-Viz, an interactive 3D visualization of a small GPT. bbycroft.net/llm
- **Cho et al.** — Transformer Explainer: interactive learning of GPT-2 in the browser. poloclub.github.io/transformer-explainer
- **Vig** — BertViz: a tool for visualizing attention in transformer models.
- **MLC team** — WebLLM / MLC-LLM: in-browser LLM inference via WebGPU. webllm.mlc.ai
- **Bai, Kolter, Koltun** — Deep Equilibrium Models, NeurIPS 2019.
- **Ramsauer et al.** — Hopfield Networks is All You Need, ICLR 2021.
- **Microsoft** — Phi-3 Technical Report (Phi-3-mini-4k-instruct).
- **W3C GPU for the Web Working Group** — WebGPU specification.

Draft v0.1. Built from the committed engine, `METHODS.md` tolerances, and the `PREDICTIONS.md` / `tests/results/` E45 artifacts; every documentation number is gated against source by `npm run verify`. To be converted to LaTeX before submission to a visualization (VIS/EuroVis), ML-systems, or interpretability venue.